

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: Sharp
Firmware Version: N/A
Model(s): 4T-B60CJ1U, 4T-B70CJ1U, 4T-B80CJ1U

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.14.0000-b007	2.13.0

Version History

Module Version	Date	Notes
1_0_2_0	10/20/2021	Fixed device connector type. Added communication sheet notes for authentication.
1_0_0_0	6/3/2020	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`
- A command pacing of 0.2 seconds is recommended between commands.
- It is recommended to wait up to 10 seconds for a response when a command is used.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='4T-B60CJ1U')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='4T-B60CJ1U')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 10002, Model='4T-B60CJ1U')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Normal' 'Full' 'Dot by Dot'	Value 'Zoom 14:9' 'Cinema 16:9' 'Overscan'	Value 'Panorama' 'Cinema 14:9'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command ATVChannelDirectCommand ¹	Value String		
# ATVChannelDirectCommand example InterfaceName.Set('ATVChannelDirectCommand', '1')			
Command AVMode	Value 'Standard' 'Dynamic'	Value 'Movie' 'Dynamic (Fixed)'	Value 'User'
# AVMode example InterfaceName.Set('AVMode', 'Standard')			
Command ChannelStep	Value 'Up'	Value 'Down'	
# ChannelStep example InterfaceName.Set('ChannelStep', 'Up')			
Command ClosedCaption	Value None		
# ClosedCaption example InterfaceName.Set('ClosedCaption', None)			
Command DTVChannelAirCommand ²	Value String		
# DTVChannelAirCommand example InterfaceName.Set('DTVChannelAirCommand', None)			
Command DTVChannelCable1Command ³	Value String		
# DTVChannelCable1Command example InterfaceName.Set('DTVChannelCable1Command', '1.1')			
Command DTVChannelCable2Command ⁴	Value String		
# DTVChannelCable2Command example InterfaceName.Set('DTVChannelCable2Command', '1')			
Command Input	Value 'TV' 'Video'	Value 'HDMI 1' 'USB'	Value 'HDMI 2' 'Home Network'
# Input example InterfaceName.Set('Input', 'TV')			

Global Scripter Module Communication Sheet

Command Keypad	Value '0' – '9'	Value 'Dot'	
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Menu' 'Left' 'Return'	Value 'Up' 'Right' 'Exit'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command Mute	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Set('Mute', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

¹String value varies from 1-135 when the TV is set to Cable, and from 2-69 when the TV is set to Air.

²String value varies from 100-9999.

³The input value for this command should be set as XXX.YYY. String value varies from 1-999 for XXX and 0-999 for YY, where XXX is the major channel and YY is the minor channel.

⁴String value varies from 0-6383 or 10000-19999.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})  
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})  
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},  
FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)  
Value = InterfaceName.ReadStatus(Command)  
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)  
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Normal' 'Full' 'Dot by Dot'	Value 'Zoom 14:9' 'Cinema 16:9' 'Overscan'	Value 'Panorama' 'Cinema 14:9'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AVMode	Value 'Standard' 'Dynamic'	Value 'Movie' 'Dynamic (Fixed)'	Value 'User'
# AVMode example InterfaceName.Update('AVMode') Value = InterfaceName.ReadStatus('AVMode') InterfaceName.SubscribeStatus('AVMode', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Input	Value 'TV' 'Video'	Value 'HDMI 1' 'USB'	Value 'HDMI 2' 'Home Network'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command Mute	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Update('Mute') Value = InterfaceName.ReadStatus('Mute') InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Global Scripter Module Communication Sheet

Command	Value
Volume	0 to 100 in steps of 1
<pre># Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

Cable and Adapter Requirements

Captive Screw to TRS connector

Notes for the Device

To ensure the functionality, please set the following:

- Menu → Initial Setup → Quick Start Mode → On
- Menu → Initial Setup → Easy Setup → Power Saving → No

Serial communication

Port Type: RS-232

Baud Rate: 115200

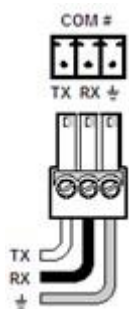
Data Bits: 8

Parity: None

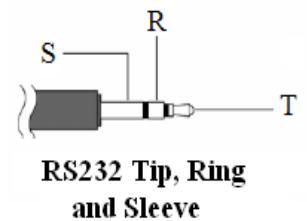
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		T	TxD
RxD		R	RxD
GND		S	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 10002

Logon Credentials Supported: Yes

Multi-Connection Capabilities: Undetermined

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To ensure the functionality, please set the following:

- Menu → Initial Setup → Quick Start Mode → On
- Menu → Initial Setup → Easy Setup → Power Saving → No
- Menu → Network Settings → Wake on Wi-Fi/LAN → Off
- Menu → Network Settings → IP Control
- Enable authentication by setting a username and password on the device, and supply the credentials to the driver in GS.

Appendix A. Set Commands

Aspect Ratio Cinema 14:9	WIDE6 \x0D
Aspect Ratio Cinema 16:9	WIDE5 \x0D
Aspect Ratio Dot by Dot	WIDE10 \x0D
Aspect Ratio Full	WIDE4 \x0D
Aspect Ratio Normal	WIDE1 \x0D
Aspect Ratio Overscan	WIDE14 \x0D
Aspect Ratio Panorama	WIDE3 \x0D
Aspect Ratio Zoom 14:9	WIDE2 \x0D
ATV Channel Direct Command None	DCCH1 \x0D
AV Mode Dynamic	AVMD5 \x0D
AV Mode Dynamic (Fixed)	AVMD6 \x0D
AV Mode Movie	AVMD2 \x0D
AV Mode Standard	AVMD1 \x0D
AV Mode User	AVMD4 \x0D
Channel Step Down	CHDW \x0D
Channel Step Up	CHUP \x0D
Closed Caption None	CLCP0 \x0D
DTV Channel Air Command 1	DA2P0001\x0D
DTV Channel Cable 2 Command 1.1	DC2U001 \x0D DC2L001 \x0D
DTV Channel Cable 2 Command 1	DC100001\x0D
Input HDMI 1	IAVD1 \x0D
Input HDMI 2	IAVD2 \x0D
Input Home Network	IAVD5 \x0D
Input TV	IAVD0 \x0D
Input USB	IAVD4 \x0D
Input Video	IAVD3 \x0D
Keypad 0	RCKY0 \x0D
Keypad 9	RCKY9 \x0D
Keypad Dot	RCKY10 \x0D
Menu Navigation Down	RCKY42 \x0D
Menu Navigation Enter	RCKY40 \x0D
Menu Navigation Exit	RCKY46 \x0D
Menu Navigation Left	RCKY43 \x0D
Menu Navigation Menu	RCKY38 \x0D
Menu Navigation Return	RCKY45 \x0D
Menu Navigation Right	RCKY44 \x0D
Menu Navigation Up	RCKY41 \x0D
Mute Off	MUTE2 \x0D
Mute On	MUTE1 \x0D
Power Off	POWR0 \x0D
Power On	POWR1 \x0D
Volume 0	VOLM0 \x0D

Global Scriptor Module Communication Sheet

Revision: 10/20/2021

Volume 100

VOLM100 \x0D

Appendix B. Update Commands

Aspect Ratio	WIDE????\x0D
AV Mode	AVMD????\x0D
Input	IAVD????\x0D
Mute	MUTE????\x0D
Power	POWR????\x0D
Volume	VOLM????\x0D